Experiments: Part II
Introduction to Some Advanced Topics

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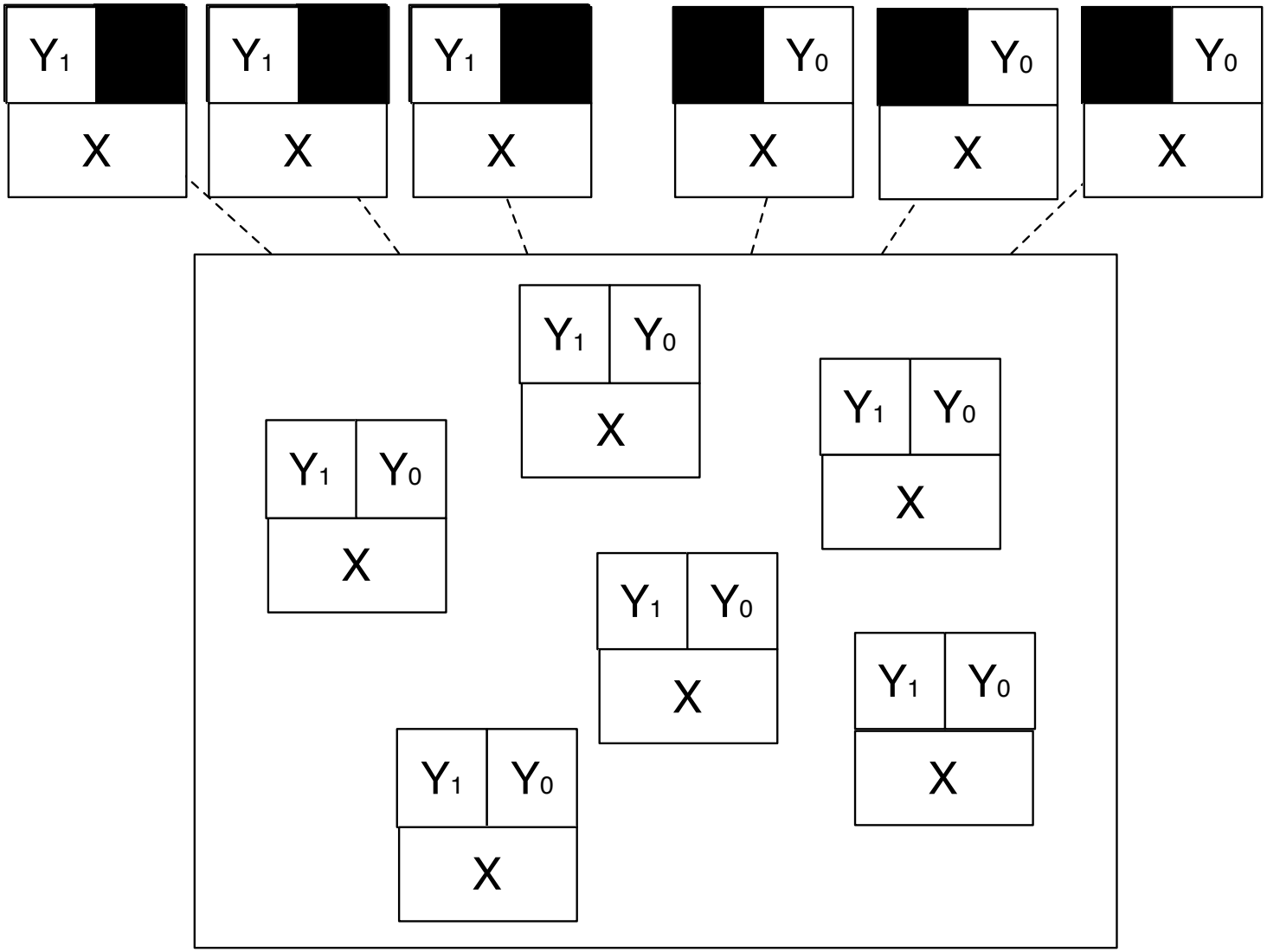
Some Important Topics in Experimental Design and Analysis

A brief introduction to some common topics:

- Covariate adjustment
- Blocking
- Emphasizing design stage
- Ethics

Covariates and Experiments



Covariates

- Unlike potential outcomes, you observe baseline covariates for all units. We mean "pre-treatment" covariates here, which are those whose values are predetermined wrt D_i/ do not depend on D_i.
- Randomization implies *covariate balance*,

$$p(X|D = 1) = p(X|D = 0)$$

- If this is not the case, then one of two possibilities:
  - randomization was compromised.
  - unlucky sampling
- Good idea to test for covariate balance on important covariates, using "balance tests" (eg. t-tests, F-tests, etc.)

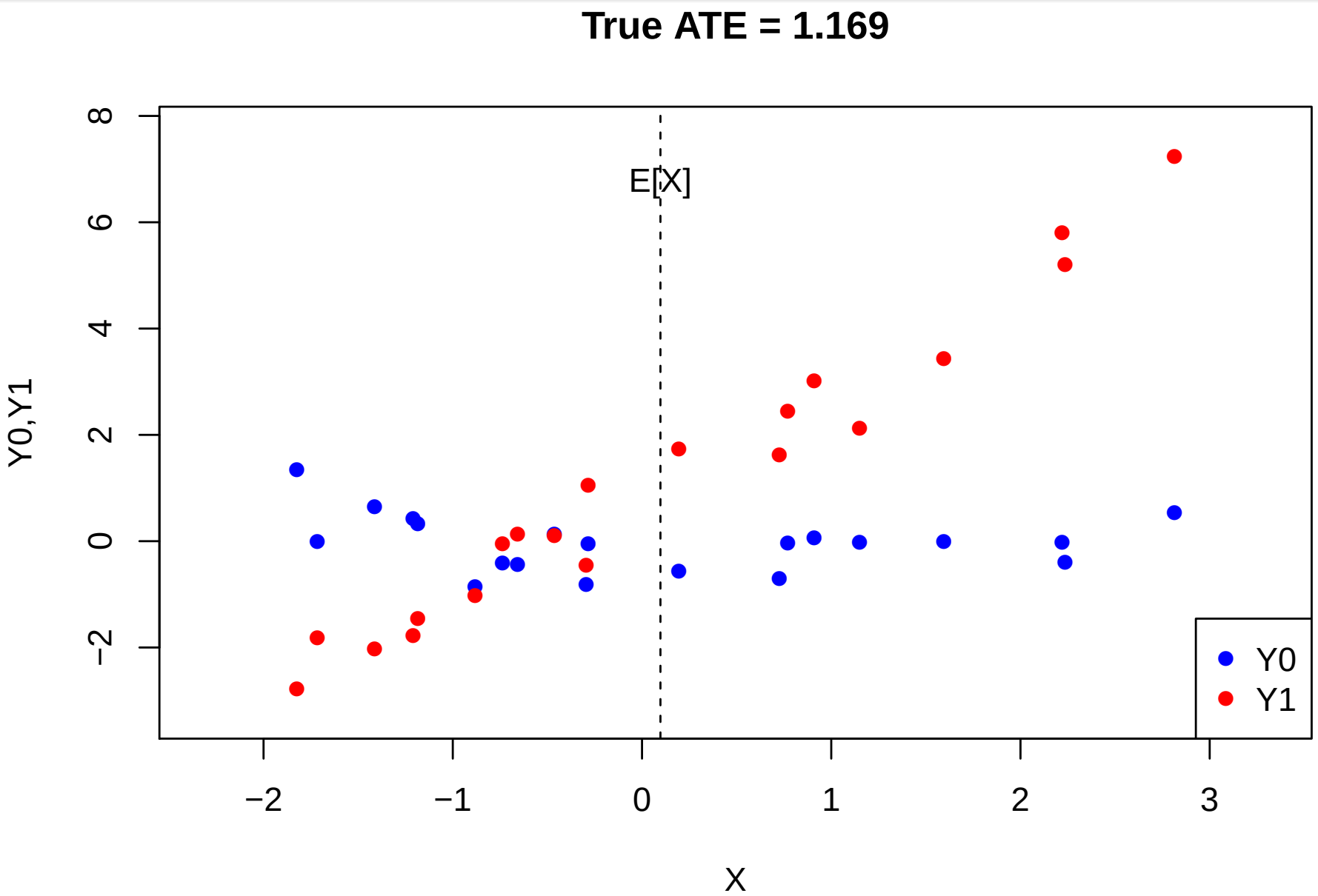
Regression with Covariates

- A common model with experimental data:

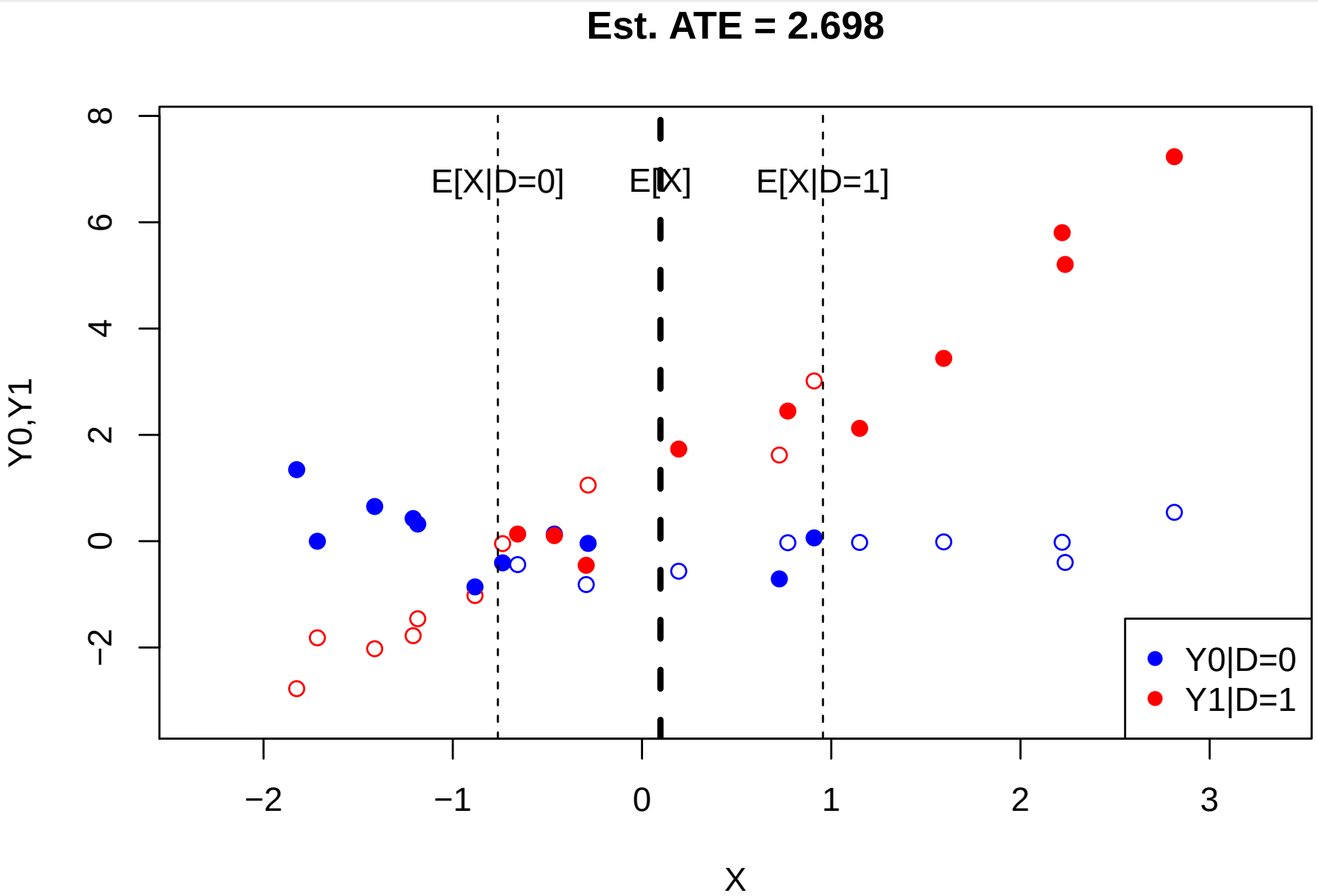
$$Y_i = \alpha + \tau D_i + X_i\beta + \varepsilon_i$$

- But why include X_i when experiments "control" for covariates by design?
  - Correct chance imbalances that may push $\hat{\tau}$ far from τ_{ATE}.
  - Increase precision: How does this happen?
- ATE estimates are robust to model specification (with sufficient N).
- But, do not control for **post-treatment** covariates!

Regression Adjustment For Experiments

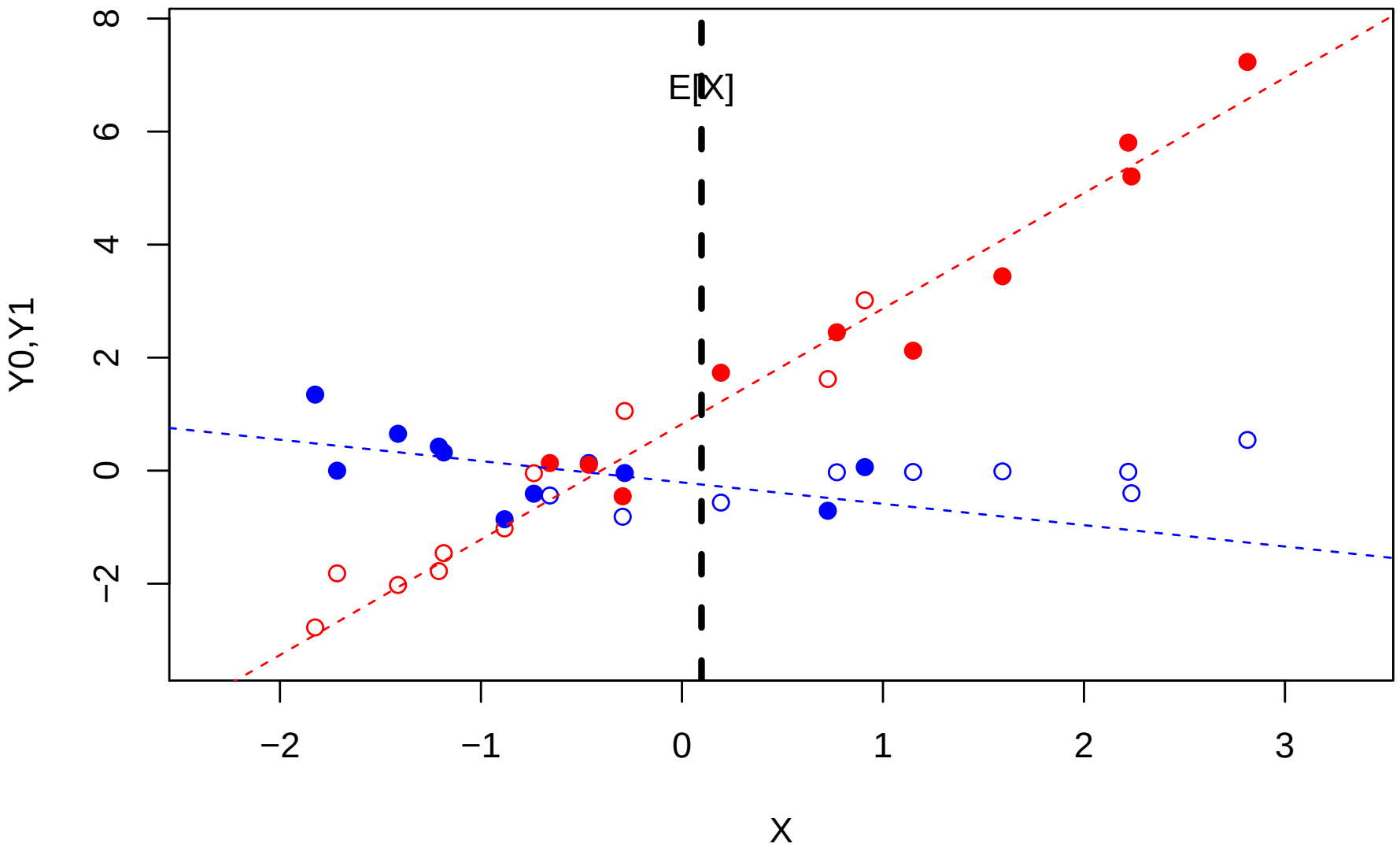


Regression Adjustment For Experiments

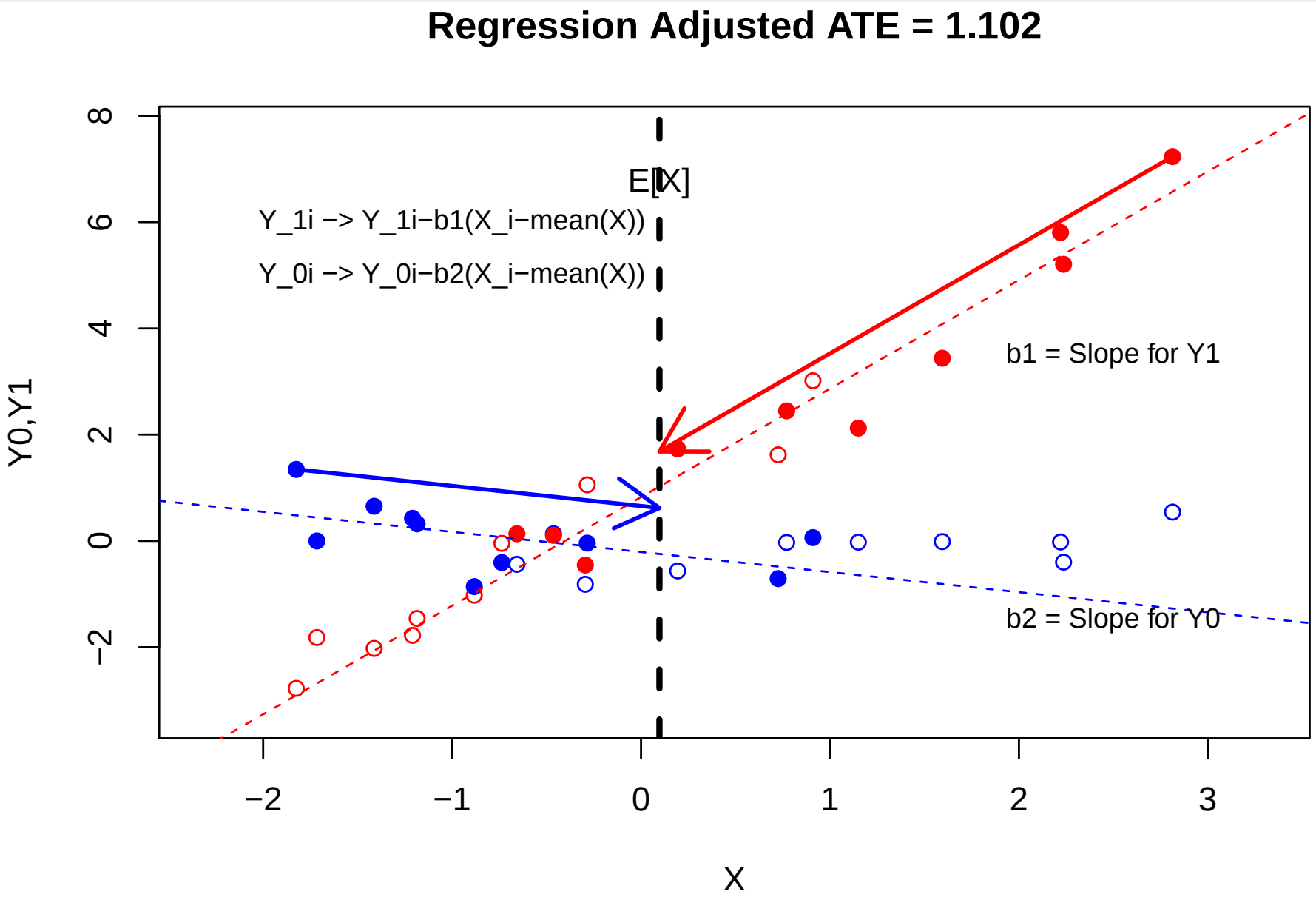


Regression Adjustment For Experiments

Consider regression lines (interacting with D)



Regression Adjustment For Experiments



Covariate Adjustment with Regression

Ever skeptical, Freedman (2008) argued that regression of the form:

$$Y_i = \alpha + \tau_{reg} D_i + \beta_1 X_i + \varepsilon_i$$

- $\hat{\tau}_{reg}$; small bias on order of $1/N$
- $\hat{\tau}_{reg}$ will not necessarily improve precision, confidence intervals wrong.

Lin (2013) replies, showing:

- in sufficient samples, these problems are minimal
- robust SEs give asymptotically consistent or conservative CIs
- if you interact covariates with treatment, it never hurts asymptotic precision, likely increases precision:

$$Y_i = \alpha + \tau_{interact} D_i + \beta_1 \cdot (X_i - \bar{X}) + \beta_2 \cdot D_i \cdot (X_i - \bar{X}) + \varepsilon_i$$

- this is what we did graphically
- interactions not needed in large sample, but might as well

Why specify the interactive model this way?

Consider our model,

$$Y_i = \alpha + \tau_{interact} D_i + \beta_1 \cdot (X_i - \bar{X}) + \beta_2 \cdot D_i \cdot (X_i - \bar{X}) + \varepsilon_i$$

Review:

- What is $\frac{\partial Y_i}{\partial X_i}$? $\frac{\partial Y_i}{\partial D_i}$?

Interpreting $\tau_{interact}$:

- What is $\mathbb{E}[Y_i | D_i = 0, X_i = \overline{X}]$?
- What is $\mathbb{E}[Y_i | D_i = 1, X_i = \overline{X}]$?
- What is $\mathbb{E}[Y_i | D_i = 1, X_i = \overline{X}] - \mathbb{E}[Y_i | D_i = 0, X_i = \overline{X}]$?
- What is $\mathbb{E}[Y_i | D_i = 1, X_i = x^*] - \mathbb{E}[Y_i | D_i = 0, X_i = x^*]$ for $x^* \neq \overline{X}$?

So,

- $\tau_{interact}$ tells you comparison after "moving" all points to $X_i = \overline{X}$
- same as $\mathbb{E}_X[\hat{\tau} | X = x]$, so convenient (faster: what is $\mathbb{E}_X[\frac{\partial Y_i}{\partial D_i}]$?)
- model also gives you $ATE(X)$ for any $x \in X$
- allowing the interaction is a good idea, emerging as standard practice

Why are Experimental Findings Robust to Alternative Specifications?

Two facts:

1. *any* bivariate regression of some Y on X has $\hat{\beta} = \frac{\widehat{\text{Cov}}(Y,X)}{\widehat{\text{Var}}(X)}$.
2. the coefficient on the kth variable, X_k, from a multivariate regression can be obtained by (i) regress X_k on other X's, call residual $\tilde{X}_k$.(ii) regress Y on $\tilde{X}_k$. (Or $\tilde{Y}$ on $\tilde{X}_k$ for similarly defined $\tilde{Y}$)

All together, this gives the Frisch-Waugh-Lovell (FWL) theorem:

$$\beta_k = \frac{Cov(\tilde{Y}_i, \tilde{X}_{ki})}{Var(\tilde{X}_{ki})}$$

Where,

- $\tilde{X}_{ki}$ is the residual from a regression of X_{ki} on all other covariates
- $\tilde{Y}_i$ may have all other X partialled out as well or you can use the original Y_i.

Why are Experimental Findings Robust to Alternative Specifications?

If our "X_k" of interest is the D:

$$\hat{\beta}_D = \frac{\widehat{\mathrm{Cov}}(\tilde{Y}_i, \tilde{D}_i)}{\widehat{\mathrm{Var}}(\tilde{D}_i)}$$

where $\tilde{D}_i$ is the residual part of D_i after regressing on X_i

- For experimental data, on average, what will $\tilde{D}_i$ look like?
- What does this suggest about the effect of adding covariates to the simple regression of Y on D?

Summary: Covariate Adjustment with Regression

- "Agnostic" view of regression: one does not need to believe in the classical linear model (linearity and constant treatment effects) to advocate OLS covariate adjustment in randomized experiments
- Helpful both to protect against chance imbalance and improve efficiency.
- Since covariates are balanced by design, results typically not model dependent
- Best reason to be skeptical of adjustment: fishing.
  - always report unadjusted result first
  - best if adjustment strategy pre-specified

Blocking

- Prior to randomization, collect background data on each unit.
- Why leave it to chance to balance on these, especially in small sample?
- Blocking: pre-stratify sample, randomize within each stratum to ensure groups start out with identical distribution of the blocked factors.
- You effectively run a separate experiment within each stratum; randomization will balance unobserved attributes
- Why is this helpful?
  - Four subjects with Y_{0i} given by $\{2, 2, 8, 8\}$. (Say Y_{1i} are similar).
  - Simple random assignment into two treated/control will place $\{2, 2\}$ and $\{8, 8\}$ together in the same group one-third of the time

Blocking

Imagine you run an experiment where you block on gender.

Obtain block specific ATEs, $\tau_f = \mathbb{E}[\tau|f]$, $\tau_m = \mathbb{E}[\tau|m]$.

(Let's review law of iterated expectations)

Applying the LIE,

$$\mathbb{E}[\tau] = p[f]\mathbb{E}[\tau|f] + p[m]\mathbb{E}[\tau|m]$$

The sample analog is unbiased for this:

$$\hat{\tau}_B = \frac{N_f}{N_f + N_m} \cdot \hat{\tau}_f + \frac{N_m}{N_f + N_m} \cdot \hat{\tau}_m$$

or more generally, if there are J strata or blocks, then

$$\hat{\tau}_B = \sum_{j=1}^{J} \frac{N_j}{N} \hat{\tau}_j$$

Blocking

Taking the number of units per block as fixed (i.e. conditional on the experimental design), the variance is:

$$\mathrm{Var}(\hat{\tau}_B) = \left(\frac{N_f}{N_f + N_m}\right)^2 \mathrm{Var}(\hat{\tau}_f) + \left(\frac{N_m}{N_f + N_m}\right)^2 \mathrm{Var}(\hat{\tau}_m)$$

or more generally

$$Var(\hat{\tau}_B) = \sum_{j=1}^{J} \left(\frac{N_j}{N}\right)^2 \mathrm{Var}(\hat{\tau}_j)$$

Blocking with Regression

Can we treat these blocks as regressors in OLS? Yes, with caution

If probability of treatment is equal across the blocks, then OLS with block "fixed effects" will estimate the ATE:

$$Y_i = \beta_0 + \tau D_i + \sum_{j=2}^{J} \beta_j \cdot B_{ij} + \varepsilon_i$$

where B_j is a dummy for the j-th block (one omitted as reference category).

If (empirical) $Pr(D = 1)$ changes by block, then OLS will put different weight on different blocks.

To correct for this, weight each observation by something that makes probabilities the same. Let p_{ij} be $Pr(D_i = 1|$ block for unit i):

$$w_{ij} = \left(\frac{1}{p_{ij}}\right) D_i + \left(\frac{1}{1 - p_{ij}}\right)(1 - D_i)$$

When Does Blocking Help?

Imagine models for complete and block-randomized designs:

$$Y_i = \alpha + \tau_{CR} D_i + \varepsilon_i \quad (1)$$

$$Y_i = \alpha + \tau_{BR} D_i + \sum_{j=2}^{J} \beta_j B_{ij} + \varepsilon_i^* \quad (2)$$

where B_j is a dummy for the j-th block. Then given iid sampling:

$$Var[\widehat{\tau}_{CR}] = \frac{\sigma_\varepsilon^2}{\sum_{i=1}^n (D_i - \bar{D})^2} \quad \text{with } \widehat{\sigma}_\varepsilon^2 = \frac{\sum_{i=1}^n \widehat{\varepsilon}_i^2}{n-2} = \frac{SSR_{\widehat{\varepsilon}}}{n-2}$$

$$Var[\widehat{\tau}_{BR}] = \frac{\sigma_{\varepsilon^*}^2}{\sum_{i=1}^n (D_i - \bar{D})^2 (1 - R_j^2)} \quad \text{with } \widehat{\sigma}_{\varepsilon^*}{}^2 = \frac{\sum_{i=1}^n \widehat{\varepsilon}_i^{*2}}{n-k-1} = \frac{SSR_{\widehat{\varepsilon}^*}}{n-k-1}$$

where R_j^2 is R^2 from regression of D on all B_j variables and a constant.

When Does Blocking Help?

$$Var[\widehat{\tau}_{CR}] = \frac{\sigma^2_\varepsilon}{\sum_{i=1}^n (D_i - \bar{D})^2} \quad \text{with } \widehat{\sigma}^2_\varepsilon = \frac{\sum_{i=1}^n \widehat{\varepsilon}_i^2}{n-2} = \frac{SSR_{\widehat{\varepsilon}}}{n-2}$$

$$Var[\widehat{\tau}_{BR}] = \frac{\sigma^2_{\varepsilon^*}}{\sum_{i=1}^n (D_i - \bar{D})^2 (1 - R_j^2)} \quad \text{with } \widehat{\sigma}^2_{\varepsilon^*} = \frac{\sum_{i=1}^n \widehat{\varepsilon}_i^{*2}}{n-k-1} = \frac{SSR_{\widehat{\varepsilon^*}}}{n-k-1}$$

where R_j^2 is R^2 from regression of D on all B_j variables and a constant.

- If R_j^2 is high, variance inflated. But what can we say about R_{ij}^2 when we're talking about blocking?
- If blocks explain outcomes strongly, how will $\widehat{\sigma}^2_{\varepsilon^*}$ compare to $\widehat{\sigma}^2_\varepsilon$?
- Adding DOF adjustment, we end up asking whether $\frac{SSR_{\widehat{\varepsilon^*}}}{n-k-1} < \frac{SSR_{\widehat{\varepsilon}}}{n-2}$

Fair Trade Labeling Experiment (Hainmueller et al, 2012)

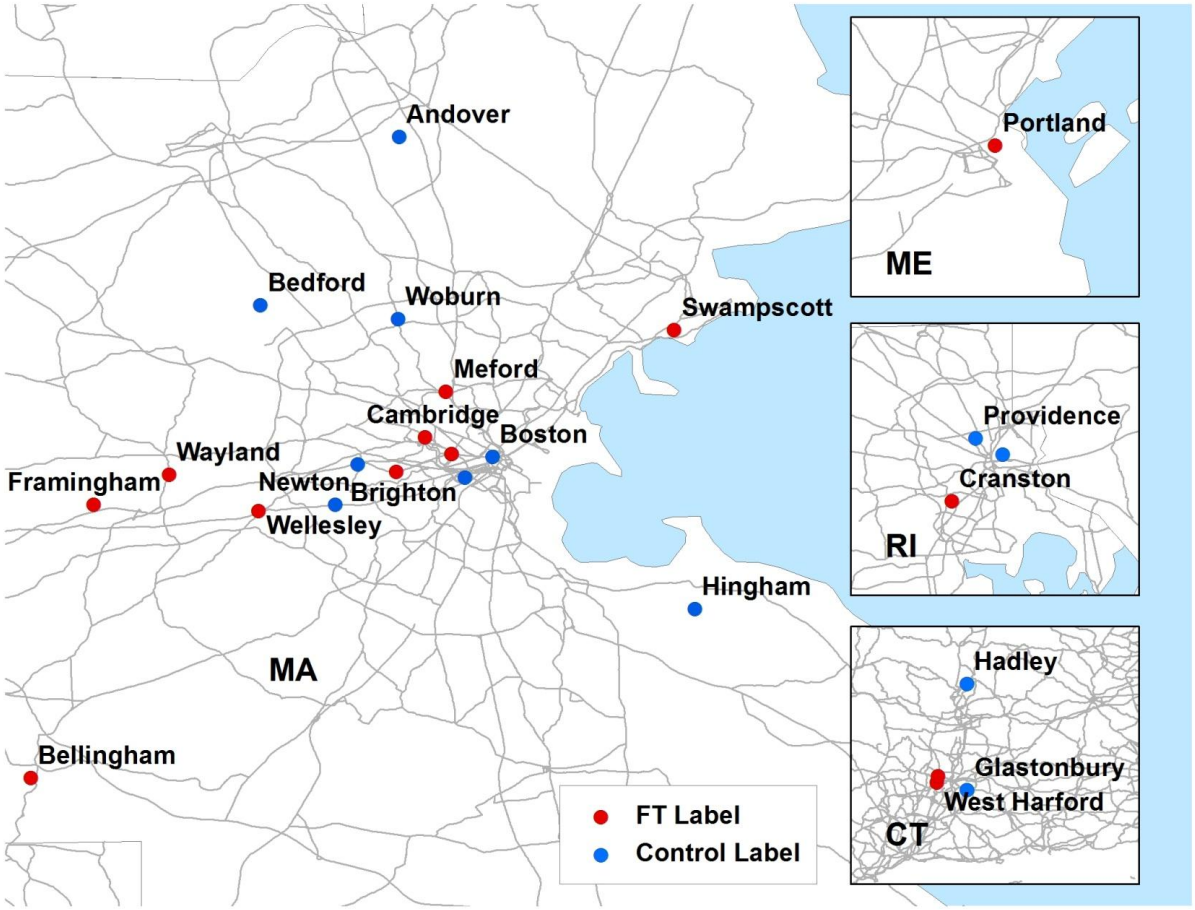
Label Experiment

Treatment

Control

Example: Fair Trade Labeling Experiment

Matched Pairs: Phase 1



Example: Fair Trade Labeling Experiment

```
> d <- read.dta("FTdata.dta")
> head(d)
  store pair FTweek lnsalesd
1     1    1      1     3.20
2     4    1      0     2.77
3     6    2      1     4.18
4     9    2      0     4.04
5    21    3      1     4.30
6    24    3      0     3.93
```

Example: Fair Trade Labeling Experiment

```
> cr.out <- lm(lnsalesd~FTweek,data=d)
> coeftest(cr.out,vcov = vcovHC(cr.out, type = "HC1"))

t test of coefficients:

            Estimate Std. Error t value Pr(>|t|)
(Intercept)  4.35000    0.16079 27.0537   <2e-16 ***
FTweek       0.12385    0.21424  0.5781   0.5686
--
```

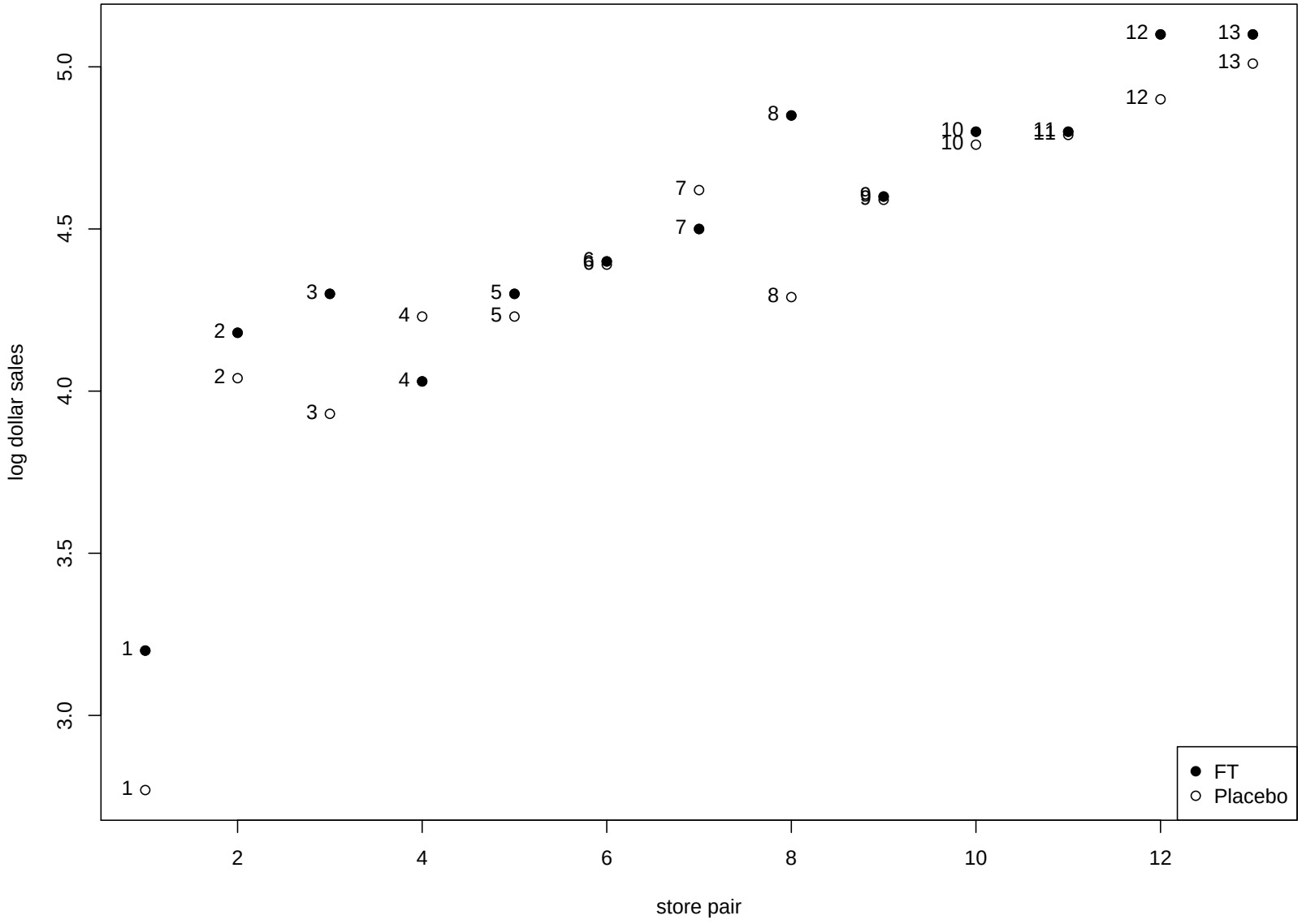
Example: Fair Trade Labeling Experiment

```
> br.out <- lm(lnsalesd~FTweek+as.factor(pair),data=d)
> coeftest(br.out,vcov = vcovHC(br.out, type = "HC1"))

t test of coefficients:

                   Estimate Std. Error t value  Pr(>|t|)
(Intercept)        2.923077   0.162144 18.0277 4.671e-10 ***
FTweek             0.123846   0.060176  2.0581 0.0619840
as.factor(pair)2   1.125000   0.159549  7.0511 1.335e-05 ***
as.factor(pair)3   1.130000   0.204440  5.5273 0.0001304 ***
as.factor(pair)4   1.145000   0.231925  4.9369 0.0003439 ***
as.factor(pair)5   1.280000   0.161773  7.9123 4.208e-06 ***
as.factor(pair)6   1.410000   0.169987  8.2948 2.591e-06 ***
as.factor(pair)7   1.575000   0.203689  7.7324 5.317e-06 ***
as.factor(pair)8   1.585000   0.277319  5.7154 9.675e-05 ***
as.factor(pair)9   1.610000   0.169987  9.4713 6.420e-07 ***
as.factor(pair)10  1.795000   0.165195 10.8660 1.450e-07 ***
as.factor(pair)11  1.810000   0.169987 10.6479 1.810e-07 ***
as.factor(pair)12  2.015000   0.164183 12.2729 3.763e-08 ***
as.factor(pair)13  2.070000   0.160298 12.9134 2.127e-08 ***
--
```

Example: Fair Trade Labeling Experiment



How much is explained by blocks?

```
> summary(lm(lnsalesd~as.factor(pair),data=d))
Coefficients:
                   Estimate Std. Error t value Pr(>|t|)
(Intercept)          2.9850     0.1212  24.621 2.72e-12 ***
as.factor(pair)2     1.1250     0.1715   6.562 1.82e-05 ***
as.factor(pair)3     1.1300     0.1715   6.591 1.74e-05 ***
as.factor(pair)4     1.1450     0.1715   6.678 1.52e-05 ***
as.factor(pair)5     1.2800     0.1715   7.466 4.73e-06 ***
as.factor(pair)6     1.4100     0.1715   8.224 1.65e-06 ***
as.factor(pair)7     1.5750     0.1715   9.186 4.77e-07 ***
as.factor(pair)8     1.5850     0.1715   9.245 4.44e-07 ***
as.factor(pair)9     1.6100     0.1715   9.390 3.71e-07 ***
as.factor(pair)10    1.7950     0.1715  10.469 1.05e-07 ***
as.factor(pair)11    1.8100     0.1715  10.557 9.56e-08 ***
as.factor(pair)12    2.0150     0.1715  11.752 2.68e-08 ***
as.factor(pair)13    2.0700     0.1715  12.073 1.94e-08 ***
--
Residual standard error: 0.1715 on 13 degrees of freedom
Multiple R-squared:  0.9474,        Adjusted R-squared:  0.8988
F-statistic:  19.5 on 12 and 13 DF,  p-value: 2.356e-06
```

Conclusions: Blocking

How does blocking help?

- Increases efficiency if the blocking variables predict outcomes
- Can help with small sample bias due to “bad” randomization
- Powerful, especially in small to medium sized samples. Can save your experiment!

What to block on?

- Two constraints: data you can get, number of units to work with.
- Irrelevant variables: burn up DOFs
- Most important: predictors of potential outcomes!
- Pre-Tx outcomes is great; add other main predictors if available/helpful
- Another consideration: variables desired for subgroup analysis

How to block?

- Stratification
- Pair-wise matching; two pair-matching

Analysis with Blocking

"**As ye randomize, so shall ye analyze**" (Senn 2004)

- Account for the method of randomization when doing analysis
- If using OLS,
  - include block dummies
  - typically use clustered SEs
  - if probability of treatment assignment varies across blocks, reweight
- If doing randomization inference, block design influences construction of Ω: re-randomization is within blocks.
- e.g., if doing bootstrap SE, use the block bootstrap (resample at level of blocks)
- Failure to account for the method of randomization in these ways can result in incorrect test size.

Hardwork in the Design and Implementation

Statistics are often easy; challenge is in implementation and design

- Finding partners, manage relationships will be common failure point
- Possible benefits to implementers
  - free “consulting” to assist with monitoring and evaluation
  - allocating limited resources (dealing with large target group)
  - encouragement designs increase uptake
  - ...but if implementers are happy with a much cheaper (non-causal) evaluation, may not be willing to do what it takes
- Iron law of field experiments: keep it simple. Many ways to fail.
- Getting difficult to publish studies that only examine the efficacy of some intervention: look for substantively interesting treatment and outcome.

Ethics and Experimentation

- Fearon, Humphreys, and Weinstein (2009) used a field experiment to examine if community-driven reconstruction programs foster social reconciliation in post-conflict Liberian villages.
- Outcome: funding raised for collective projects in public goods game played with 24 villagers. Total payout to village is publicly announced.

We received a report that leaders in one community had gathered villagers together after we left and asked people to report how much they had contributed. We moved quickly to prevent any retribution in that village, but also decided to alter the protocol for subsequent games to ensure greater protection for game participants.

These changes included stronger language about the importance of protecting anonymity, random audits of community behavior, facilitation of anonymous reporting of violations of game protocol by participants, and a new opportunity to receive supplemental funds in a postproject lottery if no reports of harassment were received.

Ethics and Experimentation

IRB approval typically required.

Key principals (Belmont Report, 1978):

- **Respect for persons**: informed consent.
  - Informed consent often done as part of the baseline survey.
  - If risks are minimal and consent will undermine the study, then informed consent rules sometimes waived.
- **Benefice**: Avoid knowingly doing harm. Does not mean that all risk can be eliminated, but possible risks must be balanced against overall benefits to society of the research.
  - Control group is often difficult: denying access to some benefit.
  - however, project benefits often finite anyway, so why not distribute them by randomization...but that won't always prevent possible envy problems
  - treatments that are occurring already in the wild pose minimal additional risk, but still may be criticized when used for research
- **Justice**: Benefits and risks should accrue to same group or individuals
  - evaluate interventions that are relevant to the subject population

FACEBOOK SHOULDN'T CHOOSE WHAT STUFF THEY SHOW US TO CONDUCT UNETHICAL PSYCHOLOGICAL RESEARCH.
THEY SHOULD ONLY MAKE THOSE DECISIONS BASED ON, UH...
HOWEVER THEY WERE DOING IT BEFORE.
WHICH WAS PROBABLY ETHICAL, RIGHT?

Conclusion: Experiments

- Random assignment is great, but can use a helping hand sometimes...
- Design: blocking to improve balance
  - improves balance
  - improves efficiency
  - can save your experiments in small samples
- Analysis: covariate adjustment
  - can improve efficiency and finite sample bias
  - relatively safe, use interactive model (Lin 2013)
  - main risk is user-defined specification (fishing)
    - report unadjusted results first
    - commit to adjustment plan, use non-arbitrary choices
    - typically results will stable though
- Beyond methodology, remember to consider:
  - can your experiment reveal an interesting effect?
  - will it have external validity
  - ethical concerns